Subject: *Database Management System 1302360*
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PLEASE NOTE THAT THIS PROJECT IS WITH A TOTAL OF 30 POINTS.

This document serves as a comprehensive guide for the project. It outlines the project description, requirements, and important instructions. The project must be described in detail using proper English grammar and formatting. The description should be clear, well organized, and should include the following phases.

I. Project Proposal and Requirement Phase (10%) – Due to Week 4

In this initial phase, students will act as systems analysts to identify a real-world problem and propose a robust database solution. This phase defines the scope and boundaries of your entire project. You are expected to clearly document what the system will do (functional requirements) and the performance, security, and reliability standards it must meet (non-functional requirements).

1. **Choose a real-world system such as:**
   - Hospital Database
   - E-commerce System
   - University Registration System
   - Library Management System
   - Banking System
2. **Problem Definition. Explain the following:**
   - The current system or scenario
   - Problems that require a database solution
   - Benefits of implementing the database
3. **System Requirements**
   - Functional Requirements such as: Add, update, delete records, generate reports, Search operations and Manage transactions.
   - Non-Functional Requirements include: Data security, Performance, Reliability and Backup and recovery.

II. Project Design and Analysis Phase (10%) – Due to Week 7

This phase transitions your project from a conceptual idea to a structured architectural blueprint. You will apply theoretical knowledge of relational algebra and storage mechanisms to design a logical and physical database schema. A strong emphasis is placed on justifying your design choices to ensure data integrity and optimal disk storage.

1. **Data Analysis. Identify the following:**
   - Entities
   - Attributes
   - Relationships
   - Estimate the data volume for future storage planning (e.g., the expected number of records per year).

2. **ER Diagram (Conceptual Design) that includes**
   - Entities
   - Attributes
   - Primary keys
   - Relationships
   - Cardinalities

3. **Schema Logical Design. Convert ERD into:**
   - Tables
   - Keys
   - Constraints

4. **Relational Algebra (ILO: S1):**
   - Construct database queries using relational algebra operations.
   - Define at least 5 operations using relational algebra, such as Selection (σ), Projection (π), Join (⋈), Union (∪), and Difference (−). Selection (σ)

5. **Disk Storage Planning (ILO: K2):**
   - Analyze how the database will be stored on disk.
   - Include estimated record size, number of records, and total storage requirements.
   - Select and justify a file organization method: Heap file, Sequential file, or Hash file (**Provide a justification for each method and explain why it was selected).**

6. **Indexing Strategy (ILO: K1):**
   - Design indexes for performance improvement.
   - Specify Primary index, Secondary index, and Composite index.
   - Justify why each index is created and specify which queries benefit from it.

III. Project Implementations, Testing, Optimization, and Presentation Phase (10%) - Due Week 13

The final phase is where your design becomes a functional system. Acting as Database Administrators, you will deploy your logical design in Oracle, enforce security and transaction protocols, and prove your system's efficiency through rigorous query optimization and constraint testing. Database Implementation.

1. **Database Implementation:**
   - Implement the database using a DBMS such as Oracle.
   - Create tables, define constraints, insert data, and create indexes.

2. **SQL Query Implementation (ILO: C1):**
   - Include a minimum of five SQL queries demonstrating basic operations, join queries, aggregation queries, and subqueries.

3. **Query Optimization.**
   - Demonstrate query execution analysis, index usage, and query rewriting strategies.

4. **Mapping Relational Algebra to SQL.**
   - Convert relational algebra queries directly into corresponding SQL queries.

5. **Testing.**
   - Functional Testing: Insert, update, delete operations, and query output correctness.
   - Constraint Testing: Primary key constraints, foreign key referential integrity, and data validation rules.
6. **Final Presentation.**
   - Present problem overview, database design, system demonstration, key queries, and challenges faced.
   - Time Allocation: 10 minutes per group.

Recommended Project Submission Structure

- Introduction
- System Analysis & Problem Definition
- ER Diagram (Conceptual Design)
- Relational Schema (Logical Design)
- Relational Algebra Queries
- Storage Planning & Indexing Strategy
- SQL Implementation & Mapping
- Query Optimization Analysis
- Conclusion & Lessons Learned

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